CRY3261-S01

1/2" Free-Field, Prepolarized, Ultra-low noise Microphone Set

Features

- **Key Specifications**

Nominal Sensitivity	354 mV/Pa
Dynamic Range	7.5 dBA to 115 dB
Frequency Range	3.15 Hz to 20 kHz ±3 dB

- **Applications**

  Anechoic chamber test
  Product self-noise
  Environmental noise test

- **Standards**

  IEC 61094 4:1995 Measurement microphones - Part 4

- **Components**

  CRY3261 1/2" Free-field Prepolarized Microphone
  CRY516 1/2" Low-noise Preamplifier

Introduction

The CRY3261-S01 is a 1/2" free field prepolarization measurement microphone and preamplifier set that is widely used in acoustic test environments such as anechoic and quiet rooms. It has extremely low self-noise, making it ideal for detecting and measuring ultra-low noise levels. Typical applications include measuring product self-noise, computer fan noise, transformer current noise, and evaluating environmental noise.

Highlights

- **Use of Ultra-low noise Free-field Microphones**

  Ultra-low noise microphone sets have very low internal noise, allowing them to conduct acoustic measurements without significantly adding to the background noise.
  Free-field microphone sets are specifically designed for measurements in environments that are free from reflections or echoes and are widely used in fields such as acoustic research, noise monitoring, and sound system testing.

- **Compatibility**

  The CRY3261-S01 microphone set uses a BNC interface coaxial cable for output and is compatible with most capture cards on the market.
  The power supply interface uses a USB Type-C port and can be widely compatible with power supply devices.

- **Calibration**

  Each CRYSOUND microphone set is calibrated at the factory using traceable calibration equipment. Calibration certificates are provided with each unit. CRYSOUND recommends recalibration at least once a year.

- **Quality & Warranty**

  All CRYSOUND microphone capsules use 3rd generation titanium diaphragms and protection grids and synthetic sapphire insulators - resulting in the most rugged and reliable measurement microphones on the market. Titanium provides superior corrosion resistance, high temperature stability, impact resistance and strength-to-mass than traditional nickel and stainless steel. All capsules are assembled in strict clean-room environments for maximum quality.

  CRYSOUND microphones and preamplifiers are supported by a 10-year warranty—offering one of the best service guarantee in the world.

Technical Specifications

Specifications

Field Type	Free-field
Sensitivity(±1.5 dB)	354 mV/Pa, -9 dB re 1V/Pa
Frequency Response	3.15 Hz to 20 kHz ±3 dB
Polarization Voltage	0 V
Dynamic Range(re.20uPa)	7.5 dBA(typical: 6.5dBA) to 115dB
Operating Temperature	-20°C to +60°C (-4°F to +140°F)
Temperature Stability	0.012 dB/°C (-10°C to +50°C) 0.008 dB/°F (+14°F to +122°F)
Static Pressure Stability	-0.01 dB/kPa
Operating Humidity Range	0 to 90%RH no condensation
Humidity Stability	< 0.1 dB (0 to 90%RH no condensation)
Pressure Equalization Vent	Rear vented
IEC 61094-4 Type	WS2F
Output Impedance	＜20 Ω
Maximum Output Voltage	＞5.0 Vrms
DC Bias Voltage	5 V
CRY575L Interface Type	DB9 (Input) BNC (Output)

Frequency Response

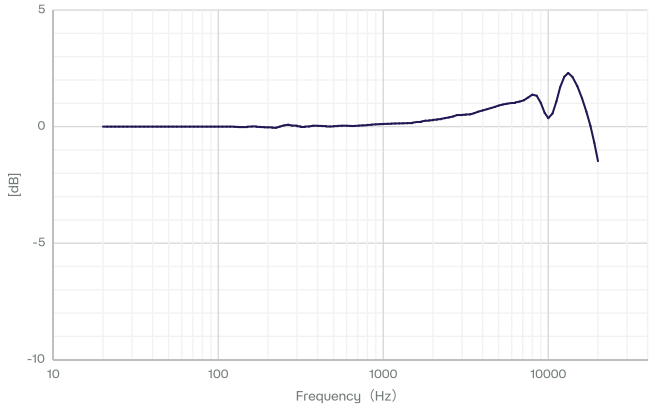

Fig.1 CRY3261-S01 Microphone Set Typical Frequency Response

Dimensions

Height with Grid	85 mm (3.346")
Diameter with Grid	13.2 mm (0.52")

Drawings(mm)[inch]

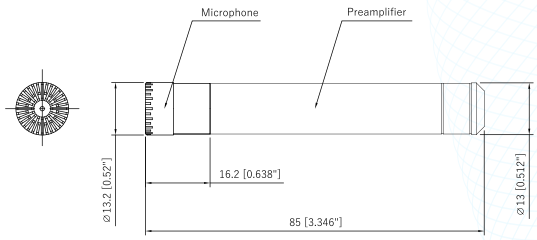

Fig.2 CRY3261-S01 Microphone Set Drawings

Ordering Information

Consisting of

Measurement Microphone	CRY3261 1/2" Free-field Prepolarized Microphone
Preamplifier	CRY516 Low-noise Preamplifier
Power Supply	CRY575L Two-channel Microphone Power Supply
Cable	BL5008 aircraft 4-core bending connector to DB9 / 1.6m

Optional Accessories

Electrostatic Actuator	TA0302 1/2" Electrostatic Actuator
Cable	BL5008 aircraft 4-Core bending connector to DB9 / 1.6m

Related Products

CRY3101-S01	1" free-field prepolarized low-noise microphone set, 50 mV/Pa, 4 Hz-16kHz, 12 dBA-135dB
CRY3201-S01	1/2" free-field prepolarized high-frequency microphone set, 12.5 mV/Pa, 3.15 Hz-40 kHz, 23 dBA-147dB
CRY3203-S02	1/2" free-field prepolarized high-sensitivity microphone set, 50 mV/Pa, 3.15 Hz-20kHz, 16 dBA-132 dB
CRY3401-S01	1/4" free-field prepolarized low-noise microphone set, 15.8 mV/Pa, 4 Hz-40 kHz, 26dB-142 dB
CRY3403-S01	1/4" free-field prepolarized high-frequency microphone set, 4 mV/Pa, 4 Hz-90kHz, 35dBA-154dB